

Kmos-0206

SUMMARY

Version: 1.0 Status: Draft

PURPOSE

The Identity Service is the authoritative source of identity, authentication, authorization, organizational structure, and accountability within the Knowledge & Media Operating System (KMOS).

Its responsibility is to establish and maintain the canonical identities of every participating actor and to ensure that every governed action performed within KMOS can be attributed to an authenticated and authorized identity.

The Identity Service SHALL establish institutional accountability.

It SHALL NOT own business data.

MISSION

Provide trustworthy identity for every participant within KMOS.

The Identity Service answers the question:

"Who performed this action, under what authority, and on whose behalf?"

RESPONSIBILITIES

The Identity Service SHALL own:

Canonical Identities Organizations Identity Lifecycle Authentication Authorization Roles Permissions Groups Service Accounts Automation Accounts AI Worker Identities Application Identities Capability Identities Connector Identities Trust Relationships Delegation Identity Audit Records Identity Registry

NON-RESPONSIBILITIES

The Identity Service SHALL NOT:

Store Knowledge Objects Store Assets Coordinate Workflows Execute Capabilities Maintain Event History Perform Search Manage Business Policies

Business authorization decisions remain policy-driven.

The Identity Service provides identity and authorization context.

CANONICAL OBJECTS OWNED

The Identity Service SHALL be the authoritative owner of:

Identity Organization Role Permission Group Authentication Credential Trust Relationship Delegation Service Account Automation Account Identity Session

Future identity objects SHALL extend this specification.

ENGINEERING PHILOSOPHY

Identity is not limited to people.

Every actor capable of participating in institutional activity SHALL possess a canonical identity.

Actors include:

Humans Organizations Applications Platform Services Capabilities Extensions AI Workers Connectors Automation Processes Future Actor Types

Identity SHALL be permanent.

Credentials MAY change.

INTERNAL MODULES

The reference implementation SHOULD contain:

Identity Registry Authentication Manager Authorization Manager Role Manager Permission Manager Organization Manager Trust Manager Delegation Manager Credential Manager Session Manager Identity API Identity Event Publisher Audit Manager

Each module SHALL have one clearly defined responsibility.

PUBLIC BUSINESS APIS

The Identity Service SHALL expose business APIs.

Examples include:

Create Identity Retrieve Identity Update Identity Deactivate Identity Create Organization Assign Role Revoke Role Grant Permission Revoke Permission Authenticate Identity Validate Session Create Service Account Create Automation Account Create AI Worker Identity Delegate Authority Retrieve Audit History

Business APIs SHALL expose identity concepts rather than authentication technologies.

CANONICAL EVENTS PUBLISHED

Examples include:

IdentityCreated IdentityUpdated IdentityDeactivated OrganizationCreated RoleAssigned RoleRevoked PermissionGranted PermissionRevoked AuthenticationSucceeded AuthenticationFailed DelegationCreated DelegationRevoked ServiceAccountCreated SessionExpired TrustRelationshipEstablished

Every identity lifecycle transition SHALL publish canonical events.

CANONICAL EVENTS CONSUMED

Examples include:

WorkflowStarted CapabilityRegistered ExtensionInstalled KnowledgeApproved PublicationReleased PolicyUpdated ConfigurationChanged

Events SHALL update accountability records where appropriate.

IDENTITY REGISTRY

The Identity Registry SHALL maintain:

Canonical Identifier Identity Type Display Name Lifecycle State Organization Roles Permissions Delegations Authentication Methods Trust Relationships Audit References

The Registry becomes the institutional directory of participants.

ORGANIZATIONS

Organizations SHALL be first-class objects.

Organizations MAY represent:

Institutions Departments Research Groups Communities Projects External Partners

Organizations SHALL define administrative boundaries without affecting canonical object ownership.

ROLES

Roles describe responsibilities rather than technical permissions.

Examples include:

Archivist Editor Publisher Researcher Reviewer System Administrator Workflow Operator Capability Developer AI Supervisor

Roles SHALL remain business-oriented.

PERMISSIONS

Permissions authorize operations.

Examples include:

Approve Knowledge Publish Assets Register Capabilities Execute Workflow Manage Organizations Review Evidence Configure Platform

Permissions SHALL remain independently governable.

DELEGATION

Authority MAY be delegated.

Delegation SHALL record:

Delegating Identity Receiving Identity Scope Duration Reason Approval

Delegation SHALL remain auditable.

AI WORKER IDENTITY

Every AI participant SHALL possess a canonical identity.

Examples include:

Speech Recognition Worker Translation Worker Knowledge Extraction Worker Editorial Assistant Story Generation Worker Rendering Worker

AI identity SHALL support:

Authentication Authorization Audit Version Tracking Capability References Human Oversight

AI SHALL never operate anonymously.

SERVICE IDENTITY

Every Platform Service SHALL possess a canonical identity.

Examples include:

Knowledge Service Asset Registry Workflow Service Event Service Capability Registry Governance Service

Services SHALL authenticate using canonical identities.

AUTHENTICATION

Authentication SHALL verify identity.

Supported categories MAY include:

Password Certificate Token Federated Identity Hardware Credential API Credential Future authentication technologies

Authentication SHALL remain implementation independent.

AUTHORIZATION

Authorization SHALL evaluate:

Identity Role Permissions Organization Delegation Policy Workflow Context Knowledge Classification Asset Classification

Authorization SHALL remain policy-driven.

SECURITY

The Identity Service SHALL support:

Credential Protection Session Management Multi-Factor Authentication Audit Logging Identity Verification Trust Management Credential Rotation

Security SHALL conform to KMOS-0190.

OBSERVABILITY

The Identity Service SHALL publish:

Authentication Metrics Authorization Metrics Session Metrics Role Statistics Organization Statistics Identity Lifecycle Metrics Audit Metrics Operational Health Institutional Accountability Metrics

SCALABILITY

The Identity Service SHALL scale independently.

Scaling dimensions include:

Organizations Identities Sessions Authentication Requests Authorization Requests Delegations Service Accounts AI Workers

Scaling SHALL remain independent of business domains.

FAILURE STRATEGY

Failures SHALL preserve institutional accountability.

Recovery SHALL support:

Credential Recovery Session Recovery Audit Preservation Identity Reconstruction Policy Re-evaluation Operational Alerts

Identity integrity SHALL remain verifiable.

TESTING STRATEGY

Testing SHALL include:

Identity Lifecycle Authentication Authorization Delegation Role Assignment Permission Evaluation Organization Management AI Identity Service Identity Performance Disaster Recovery Audit Validation

ACCEPTANCE CRITERIA

A compliant Identity Service SHALL:

Maintain canonical identities. Support organizations. Support roles. Support permissions. Support authentication. Support authorization. Support delegation. Support AI identities. Publish canonical events. Remain implementation independent.

REALITY CHECK

Capability | Existing Evidence | Status Human authentication | Existing platform authentication | Proven Service authentication | Media Pipeline infrastructure | Partial Organizational model | Existing projects | Partial AI worker identities | Concept established | Planned Capability identities | Registry concepts | Planned Delegation | Governance workflows | Early Canonical identity registry | Planned | Not Implemented Institutional trust model | KMOS specifications | Defined

The implementation SHALL prioritize human, service, and organization identities before introducing advanced identity types.

REFERENCE IMPLEMENTATION

Suggested internal architecture.

identity-service/ |—— api/ |—— application/ |—— domain/ | |—— identities/ | |—— organizations/ | |—— authentication/ | |—— authorization/ | |—— roles/ | |—— permissions/ | |—— delegation/ | |—— trust/ | |—— sessions/ |—— events/ |—— repositories/ |—— security/ |—— tests/ |—— migrations/ |—— documentation/

Authentication technology SHALL remain behind the business APIs.

CLAUDE IMPLEMENTATION GUIDE

Objective Implement the authoritative Identity Service for KMOS.

Primary Specifications KMOS-0180 KMOS-0190 KMOS-0200

Deliverables Production-ready Identity Service. Identity Registry. Authentication Manager. Authorization Engine. Organization Manager. Role & Permission Manager. Delegation Manager. Automated Tests. Deployment Manifests. Operational Documentation.

Acceptance The implementation SHALL satisfy every requirement defined in this specification. Every actor within KMOS SHALL possess a canonical identity. Identity SHALL remain independent of authentication technology.

RELATIONSHIP TO OTHER SERVICES

Knowledge Service Records ownership and stewardship.

Asset Registry Records custodianship and chain of custody.

Event Service Publishes identity lifecycle events and authorizes subscriptions.

Workflow Service Assigns human and automated tasks.

Capability Registry Registers Capability and runtime identities.

Governance Service Evaluates accountability and compliance.

Configuration Service Provides identity-related operational configuration.

The Identity Service SHALL remain the authoritative owner of institutional accountability.

ARCHITECTURAL CONSEQUENCES

Because every actor possesses a canonical identity:

Every action becomes attributable. Every approval becomes auditable. Every AI worker becomes accountable. Every service becomes trusted. Every workflow becomes explainable. Every capability becomes certifiable. Institutional accountability survives technology replacement.

The Identity Service becomes the institutional accountability engine of KMOS.

REFERENCE EXAMPLE

An editor approves a corrected transcript.

Editor authenticates. Identity is authorized. Workflow assigns approval task. Approval is completed. KnowledgeUpdated event is published. Identity is recorded within audit history. Governance records institutional accountability.

Years later, the organization can determine:

Who approved the knowledge. Under which authority. Using which workflow. Following which policy. With which supporting evidence.

Identity becomes part of institutional memory.

NEXT SPECIFICATION

KMOS-0207 defines the Governance Service.

Where:

The Knowledge Service preserves institutional meaning, The Asset Registry preserves institutional evidence, The Event Service preserves institutional communication, The Workflow Service preserves institutional coordination, The Capability Registry preserves institutional execution knowledge, The Identity Service preserves institutional accountability,

the Governance Service preserves institutional trust by ensuring that every decision, approval, certification, policy evaluation, and compliance activity is permanently recorded, explainable, and auditable.

It becomes the final foundational engine that completes the institutional core of KMOS.